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Abstract

Automated exploration of reaction networks typically assumes a single potential-energy surface. However, 3d transition-metal catalysis often involves two-state reactivity, where the rate-determining step may occur on a different spin surface. This work presents an open-source extension of AutoMeKin that incorporates spin considerations throughout. The extension features a GFN2-xTB exploration engine with spin-handling capabilities, identification of closely lying spin states, localization of minimum-energy-crossing-points (MECP) at relevant crossings, integration of MECP nodes connected by nonadiabatic edges within the reaction graph, nonadiabatic transition-state-theory rate calculations, and a multireference diagnostic based on #S2# to disable the single-determinant approach where it is not applicable. The extension is tested on three different iron systems. For H2CO and Fe(CO)5, blind path sampling was utilized, whereas targeted transition-state optimization was applied to the more complex cases. Subsequently, MECPs were localized without manual intervention for Fe(CO)5 and Tp-Fe(II)-ethyl, but not for the Fe(I) alkene-isomerization catalyst.

Keywords

automated reaction discovery, minimum-energy crossing point, multireference diagnostics, spin-forbidden reactivity, transition-metal catalysis

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Spin-Resolved Reaction-Network Exploration in AutoMeKin with Minimum-Energy-Crossing-Point Localization and an $\langle S^2 \rangle$ Single-Reference Check

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ABSTRACT

Automated exploration of reaction networks typically assumes a single potential-energy surface. However, 3d transition-metal catalysis often involves two-state reactivity, where the rate-determining step may occur on a different spin surface. This work presents an open-source extension of AutoMeKin that incorporates spin considerations throughout. The extension features a GFN2-xTB exploration engine with spin-handling capabilities, identification of closely lying spin states, localization of minimum-energy-crossing-points (MECP) at relevant crossings, integration of MECP nodes connected by nonadiabatic edges within the reaction graph, nonadiabatic transition-state-theory rate calculations, and a multireference diagnostic based on $\langle S^2 \rangle$ to disable the single-determinant approach where it is not applicable. The extension is tested on three different iron systems. For H_2CO and $Fe(CO)_5$, blind path sampling was utilized, whereas targeted transition-state optimization was applied to the more complex cases.

Subsequently, MECPs were localized without manual intervention for $Fe(CO)_5$ and Tp-Fe(II)-ethyl, but not for the Fe(I) alkene-isomerization catalyst.

1. Introduction

Automated reaction-network exploration typically assumes a single potential-energy surface, despite the prevalence of spin-state crossings in 3d transition-metal catalysis. The present work incorporates a spin-resolved component into the open-source AutoMeKin engine. Application to three iron systems demonstrates that this component can automatically detect crossings, report when the crossing energy cannot be converged, and flag nodes exhibiting multireference electronic structure. Automated chemical-reaction-network (CRN) exploration is now standard in organic and combustion chemistry and is increasingly applied to homogeneous transition-metal catalysis. AutoMeKin integrates molecular dynamics-based transition-state sampling with graph-theory network construction;[1,2] Chemoton (SCINE) provides a first-principles, gear-driven exploration approach;[3] GRRM/AFIR determines reaction pathways via artificial force induction;[4] and ReNeGate[5] and KinBot[6] focus on homogeneous catalysis and gas-phase kinetics, respectively.

The tools in question assume that there is a single potential-energy surface; in the case of 3d-metal catalysis this represents a real limitation since spin-state crossings frequently determine the mechanism.[7–9] If the search is restricted to one spin surface, it may fail to identify the crossing seam,

which provides access to the true, lower-lying transition state on a different surface. GRRM/AFIR is able to find minimum-energy seam crossings (MESX) through the use of GP/SC-AFIR,[10,11] but the software is closed source and is not incorporated into a Python-extensible graph layer that can be linked to downstream rate analysis. Chemoton[3] includes spin as part of its structure definition yet allows only a single spin state per exploration. Machine-learning methods predict transition-metal spin-state splittings and ground-state spin assignments[12,13] but do not discover crossing seams during an automated network search.

The minimum-energy crossing point (MECP) corresponds to a transition state in the case of a spin-forbidden step,[7,14] and the rate of this process is given by nonadiabatic transition-state theory using a Landau–Zener surface-hopping coefficient.[7,15] Methods are available both for *localizing* an MECP once an initial guess has been made (easyMECP,[16] ORCA 6[17]) and for *calculating the rate* thereof; the missing element has been an automated procedure which could (a) determine *where* within a network a crossing should be looked for, (b) localize the crossing, (c) incorporate it back into the network with a rate, and (d) know when the electronic structure at a node is such that it becomes too multireference for single-determinant approaches to be trusted.

This study extends AutoMeKin[1,2] to treat each spin state as a separate potential-energy surface throughout the entire workflow. The extension incorporates a multireference diagnostic that identifies nodes outside the

range of single-reference methods and demonstrates the approach on two well-established two-state systems (one stoichiometric and one catalytic), as well as a multireference test case that the diagnostic correctly flags. Collectively, these three systems illustrate the possible outcomes of the extension: a converged MECP and rate (Tp-Fe), a located crossing whose energy is not converged (Fe(I)), and a system flagged as multireference (the nitrene; Figure 1). The primary contribution is not a new CRN engine, but rather a spin-resolved layer that transforms single-surface network exploration into a multi-surface approach by automatically detecting and rate-assigning crossing nodes and incorporating an explicit single-reference check, together with the open-source release of this layer. Section 2.8 specifies which workflow components are automated for each system; Section 4 discusses the rationale for employing a semiempirical method as the spin-resolved *explorer* and DFT for spin-sensitive energetics, as well as the importance of DFT functional selection for crossing detection.

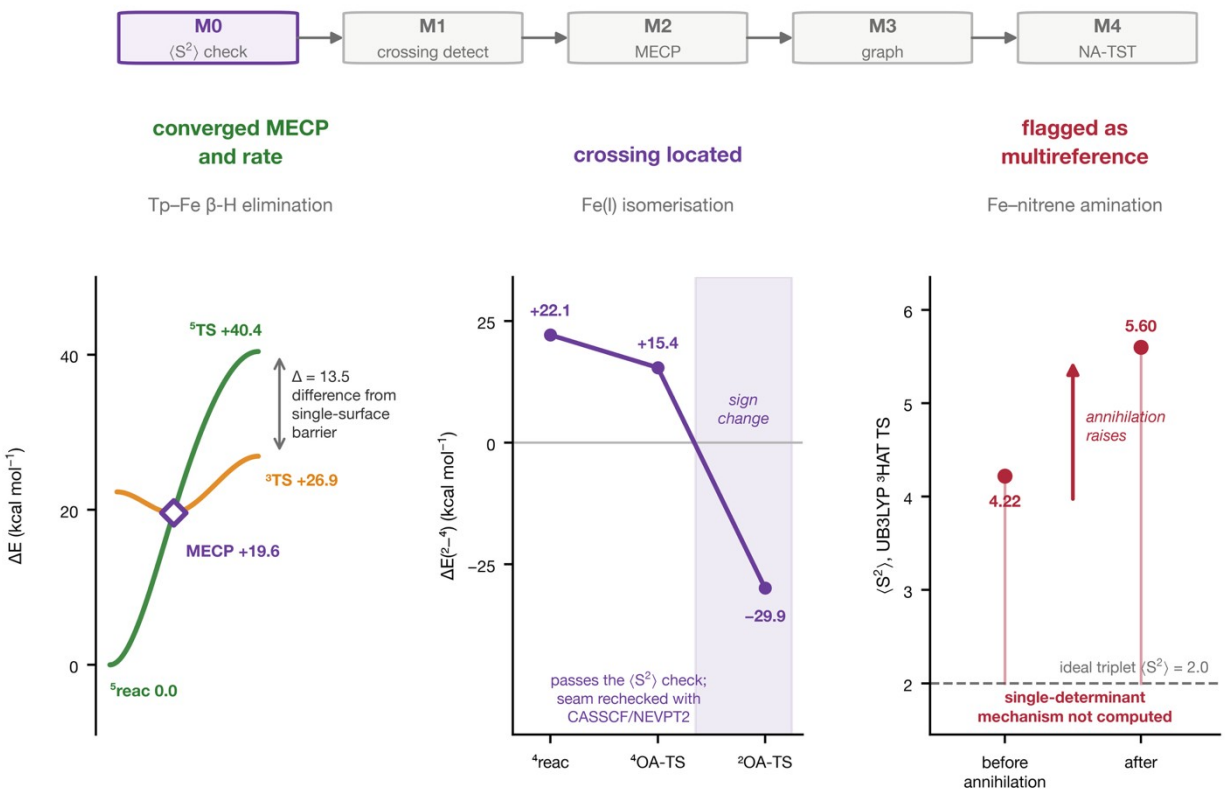


Figure 1. The three outcomes the extension can return, one per column; the three panels carry different vertical axes and are not on a common scale. The strip at the top lists the module chain (M0 ⟨S²⟩ check → M1 spin-state enumeration → M2 MECP localization → M3 graph insertion → M4 NA-TST and microkinetics), and each column below is labeled with its outcome and the system that produced it. Left, converged MECP and rate — Tp-Fe(II)-ethyl β-H elimination: relative electronic energy (ΔE, kcal/mol, UB3LYP/def2-SVP) along the β-hydride elimination coordinate on the quintet (green) and triplet (orange) surfaces, with the $^{5}/^{3}$ MECP (open purple diamond) at +19.6 lying below both the ^{3}β-H TS (+26.9) and the ^{5}β-H TS (+40.4); the 13.5 kcal/mol separation of the two transition states is the error a single-surface search would incur. Middle, crossing located, energy

not converged — Fe(I)-catalyzed alkene isomerization: vertical doublet-minus-quartet gap $\Delta E(^2-^4)$ (kcal/mol, SA-CASSCF(7,5)/def2-SVP) at the three DFT-optimized geometries (^{4}reactant, ^{4}OA-TS, ^{2}OA-TS). The $\langle S^2 \rangle$ check is passed, but the nearly degenerate $^4/^2$ seam defeats the single-determinant solver; the change of sign across the shaded region (+22.1 → +15.4 → −29.9) places the crossing in the OA-TS region. Right, flagged as multireference — Fe-porphyrin nitrene C–H amination: $\langle S^2 \rangle$ of the triplet HAT transition state at UB3LYP before and after spin annihilation (4.22 → 5.60; ideal triplet value 2.0, dashed line; Stable=Opt-confirmed). Annihilation increases the contamination rather than removing it, and does so for every functional tested, so M0 excludes the node from single-determinant treatment**.**

2. Implementation

This extension includes a spin-capable engine, a multireference diagnostic, and modules M1 to M4 (Figure 2).

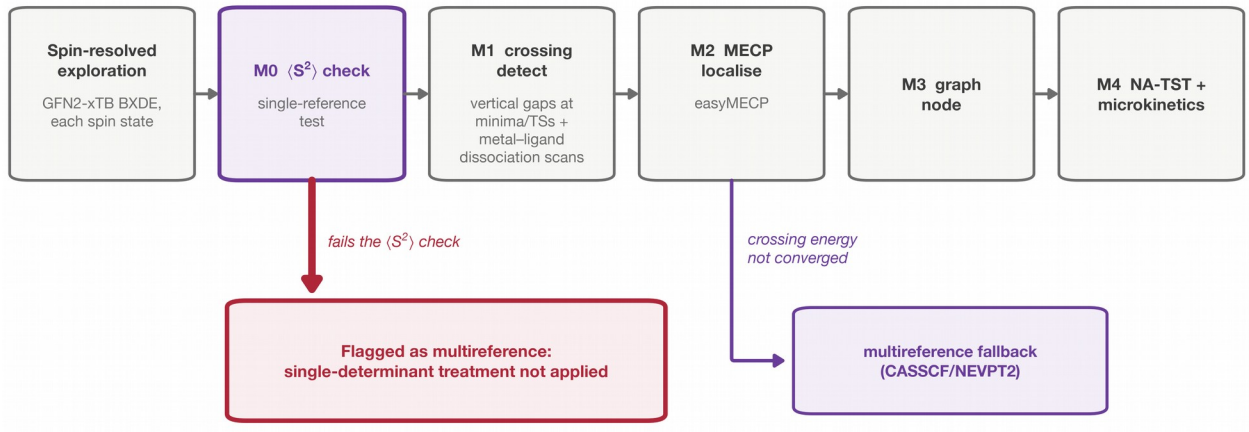


Figure 2. General organization of the spin-resolved extension. The spin-resolved GFN2-xTB exploration supplies data to the M0 $\langle S^2 \rangle$ check; those

nodes which pass go on to crossing detection (M1, including metal–ligand dissociation scans), MECP localization (M2, easyMECP), graph insertion (M3), and NA-TST/microkinetics (M4). If the check fails, the entire single-determinant branch is terminated (crimson): the node is identified as multireference and all the subsequent steps are stopped. Seams that do not converge to a single determinant at M2 are passed over to the multireference fallback (purple).

2.1 Spin-capable low-level engine

The low-level (LL) path in AutoMeKin uses PM7[18] and incorporates the molecular charge as the only information from the input deck; however, when the released code (revision 1137[2]) is examined, it is found that the spin multiplicity from the input deck is transferred to the quantum-chemistry input only at the DFT high-level (HL) stage. As a result, all low-level calculations proceed using closed-shell RHF regardless of the multiplicity that was requested, and any "different-spin" exploration in fact involves only a single potential-energy surface. For this reason we have replaced PM7 with GFN2-xTB[19] as the LL method by means of a new ASE[20]-calculator interface (XTBamk).

2.2 Low-level transition-state optimization and IRC

XTB does not have a native saddle-point optimizer or an intrinsic-reaction-coordinate (IRC) integrator, and the LL TS/IRC routines in AutoMeKin are specific to MOPAC or Entos. We have therefore included a TS/IRC layer which uses the Sella[21] ASE saddle optimizer (a first-order optimizer based

on internal coordinates) on the XTBamk calculator, the IRC being carried out by displacing along the imaginary normal mode and then carrying out a minimization on each side. Each identified saddle is verified by means of a single imaginary frequency and its connectivity is established through the two minimized IRC end points.

2.3 Multireference diagnostic (M0): ⟨S²⟩ check

The DFT and MECP procedures are only valid when the wavefunction is single-reference. Before a node is accepted, the extension carries out an ⟨S²⟩-based check: an unrestricted DFT calculation is carried out at the node's geometry for each relevant multiplicity, and ⟨S²⟩ is compared with the ideal value of $S(S+1)$ both before and after spin annihilation. The node is considered to have passed if, for each multiplicity that has been evaluated at its own optimized structure, annihilation causes ⟨S²⟩ to decrease toward $S(S+1)$ and if the deviation after annihilation $|\langle S^2\rangle_{ann} - S(S+1)|$ is less than or equal to 0.5; it is identified as multireference if annihilation increases ⟨S²⟩ or the post-annihilation deviation is greater than or equal to 1.0, and this occurs persistently across at least two functionals, so that a problem in just one functional does not lead to a false rejection; in all other cases the node is flagged for review rather than being automatically classified. In order for a node to pass, all of the functionals in the cross-check set must individually pass. In this study no intermediate cases were encountered: all the passes were within 0.4 of the ideal value after annihilation (most being within 0.005) and all the rejections were at least

2.5 units above the ideal value with annihilation causing an increase (see SI Section S2). The rule is implemented as a standalone module (m0_gate.py, included in the patch set) that extracts the $\langle S^2 \rangle$ values before and after annihilation from the unrestricted DFT output and applies these thresholds. Instead of proceeding with a flagged node, it is reported as out of scope for the single-reference DFT/MECP analysis. The critical factor is the direction in which $\langle S^2 \rangle$ changes during annihilation. The range from 0.5 to 1.0 is intentionally broad, as values within this interval are subjected to further review rather than immediate classification. Adjusting the pass limit to 10% of S(S+1) does not alter any of the reported verdicts for the 433 archived unrestricted DFT outputs. While a fixed limit of 0.5 constitutes a large fraction of S(S+1) for low-spin systems (notably two-thirds for a doublet), it is recommended that low-spin Co, Ni or Mn complexes employ a limit scaled to S(S+1). Both limits are user-configurable (see SI Section S1.2).

2.4 Spin-state enumeration and close-lying-state detection (M1)

For each node (whether it is a minimum or a transition state) in a discovered network, the neighboring spin states (those differing by ±2 in multiplicity, i.e., $\Delta S = 1$) are assessed at the node's geometry using the spin-capable LL engine, and a node is identified as a crossing candidate if the smallest vertical spin gap lies within a user-set window (default 30 kcal/mol for 3d-metal systems; SI Section S3.1). Since the LL spin-gap *calculations* are unreliable (Section 4), this window serves as a crude filter.

The *decision* and the crossing energy are based on DFT calculations, and only after the M0 check has confirmed that the node is single-reference.

Ligand-loss scan. Spin crossings may also occur along coordinates involving the dissociation of a metal–ligand bond, which are not captured by the vertical gaps at network nodes. For each minimum within 20 kcal/mol of the network's lowest minimum, M1 executes relaxed GFN2-xTB scans of each symmetry-distinct metal–ligand distance (for neutral L-type ligands) on both the base multiplicity and those differing by ±2. A channel is flagged when two curves intersect or approach each other to within the M1 window. DFT single-point calculations are then performed on the scan frames, and the frame at the DFT crossing serves as the seed for M2. If the DFT curves do not cross, the frame with the smallest gap is used instead, and M2 proceeds only if the DFT curves cross or if their minimum gap falls within the same window.

2.5 MECP localization (M2)

When a candidate that has been flagged passes the M0 check, the minimum-energy crossing point (MECP) between the two spin surfaces is determined at the DFT level by running easyMECP[16] (which is a self-contained wrapper around J. N. Harvey's MECP code[14]) from the geometry that has been flagged. In the case of nearly degenerate spin states the SCF associated with the crossing seam can be numerically unsolvable for single-determinant methods (Section 6.2); we record the various failure modes and

the procedure to address them (using relaxed seeds; falling back to a multireference approach).

2.6 MECP nodes in the network graph (M3)

The per-spin networks and the localized MECPs are combined into one spin-resolved reaction graph, with the nodes carrying their spin manifold (multiplicity), the intra-surface reactions being represented as edges, and each MECP becoming a node that is connected via nonadiabatic crossing edges to the nearest node on each surface, the connections being matched by Kabsch heavy-atom RMSD.[22,23] The graph is output in JSON format (for rate analysis) and in graphviz/mermaid format for visualization. We monitor energy provenance on a per-node basis (LL versus DFT); we compare cross-surface energies only using DFT energies calculated at a consistent level (Section 5.3 explains why consistency matters).

2.7 NA-TST rates and microkinetic post-analysis (M4)

Each accepted crossing is assigned a rate by nonadiabatic transition-state theory with a Landau–Zener double-passage surface-hopping transmission coefficient,[7]

$$k(T)=\kappa(T)\frac{k_B T}{h}\frac{Q_{MECP}}{Q_{react}}\exp\left(\frac{-\Delta E_{MECP}}{k_B T}\right)$$

$$\kappa(T)=\frac{1}{k_B T}\int_0^{\infty} p_{sh}(E_n)e^{-E_n/k_B T}\,dE_n$$

$$p_{sh}=1-P_{LZ}^2,$$

$$P_{LZ}=\exp(-2\pi\Gamma),2\pi\Gamma=\frac{2\pi H_{SO}^2}{\hbar\vee\Delta F\vee v},$$

$$v=\sqrt{2E_n/\mu}$$

with H_{SO} the spin–orbit coupling at the MECP, $|\Delta F|$ the norm of the gradient difference of the two surfaces, μ the effective mass of the hopping coordinate and E_n the kinetic energy along it. The rate is attached to the crossing edges. We then apply the energetic-span analysis (Kozuch–Shaik)[24] to the competing same-spin and spin-cross routes by folding the Landau–Zener κ into an effective barrier ($\Delta G^{\ddagger}_{cross,eff} = E_{MECP} - RT \ln \kappa$); this analysis identifies the turnover-determining transition state and whether the spin crossing is rate-limiting. The energetic-span model was formulated for catalytic cycles in which the transmission coefficient equals unity at each step. A κ that deviates from unity, here the thermally averaged Landau–Zener double-passage probability described above, is incorporated as an additive correction of $-RT \ln \kappa$ to the free energy of the relevant transition state (see Section 9 of ref. 24). For a single elementary step, such as the Tp-Fe(II)-ethyl β-H elimination, no cycle is present, and the analysis reduces to a comparison of the two effective barriers. A step list is exported for downstream full microkinetic treatment.

2.8 Scope of automation

Blind spin-resolved exploration has been carried out on H_2CO and $Fe(CO)_5$ (Section 5.1). In the case of the two iron validation complexes, blind BXDE sampling was not employed to identify the reactive steps: for the 34-atom Tp-Fe(II)-ethyl system the sampling is limited by decomposition and the β-H product is unreachable within the network, so the two per-spin β-hydride saddles were found using targeted Sella saddle optimization based on a

relaxed Cβ–H scan; the Fe(I) network was also constructed by identifying the stationary points through targeted saddle optimization on a truncated model, with reference to the previously published mechanism.[25] The complementary boundary for bimolecular encounters is given in Section 6.1. Once the saddles have been located, the spin-state enumeration (M1) and the crossing flag operate automatically on both iron systems. Automatic localization of the MECP (M2) was achieved on Tp-Fe, with a degenerate crossing being converged at the DFT level, and on $Fe(CO)_5$ from the seed of the ligand-loss scan (Section 5.1); on Fe(I) it failed, because the SCF at the near-degenerate seam is ill-conditioned for a single determinant (Section 6.2). The crossing there was instead resolved with the multireference fallback of Section 2.5 — a manually set up SA-CASSCF(7,5)+SC-NEVPT2 map — which locates the crossing but leaves its energy on the DFT scale unconverged (Section 5.3). Graph insertion (M3) and the rate and microkinetic analysis (M4) then took place automatically on both systems, with one qualification on Fe(I): the node that M3 inserts is the stored seam frame and not a converged MECP, so the rate that M4 returns depends on a crossing energy that was never converged. That rate is therefore reported as a function of the assumed crossing energy rather than as a single value (SI Section S4.7, Figure S2). The ligand-loss scan was performed on the minima of the $Fe(CO)_5$, Tp-Fe and Fe(I) networks. The number of channels at each stage is presented in Table 1, and the individual runs are detailed in SI Section S8. With the exception of the multireference treatment for the

Fe(I) crossing, all steps from M1 to M4 proceeded without manual intervention. Of the 22 MECP searches started from ligand-loss seeds (Table 1; SI Section S8.1).

Table 1. Stage at which manual input was needed, per system. Entry: how the stationary points were obtained. M1: crossings flagged at network nodes and in the ligand-loss scan (Section 2.4); for M2, the numbers of MECP searches attempted and converged or located (SI Section S8.1); for M3–M4, graph insertion and rate assignment.

System	Entry (stationary points)	M1 crossing flag	M2 MECP (attempted / converged)	M3–M4
H_2CO	blind BXDE; full network, IRC-verified TSs (one spin surface)	not applicable	not attempted	not applicable
$Fe(CO)_5$	blind BXDE, singlet and triplet; 3 (^{1}) + 6 (^{3}) TSs, each with one imaginary mode	network nodes: 0; ligand-loss scan: 16 channels flagged, 12 passed by DFT; no manual input	12 / 11; no manual input	M3: MECP inserted, no manual input; M4: not computed (no spin–orbit coupling evaluated)
Tp-Fe(II)-Et	targeted saddle optimization from a relaxed Cβ–H scan (manual); 2 per-spin β-H TSs + reactants	network nodes: 1; ligand-loss scan: 8 flagged, 8 passed by DFT; no manual input	β-H region: 1 / 1; ligand-loss: 8 / 8; no manual input	no manual input
Fe(I) alkene	targeted saddle optimization (manual); 4 stationary points	network nodes: 1; ligand-loss scan: 2 flagged, 2 passed by DFT; no manual input	1 / 0, crossing located with CASSCF/NEVPT2 set up by hand (manual); ligand-loss: 2 / 2	M3: seam frame inserted; rate reported as a function of the crossing energy (SI Section S4.7)
Fe-nitrene	targeted (Section 6.1); 2 HAT TSs	not attempted: flagged as multireference by M0	not attempted	not attempted

3. Computational details

Low level (exploration): GFN2-xTB[19] (xtb 6.7.1) with native spin control (--uhf).

Multireference diagnostic (M0) involves unrestricted DFT $\langle S^2 \rangle$ calculations before and after annihilation, and this is verified using UB3LYP[26–28], OPBE[29–31], and TPSSh[32,33] (the flag has to remain consistent across the different functionals).

High level (energetics), Tp-Fe system: UB3LYP/def2-SVP[34] was used for the reference network; single-point energy calculations were carried out using UB3LYP/def2-TZVP and UB3LYP/def2-TZVP+D3(BJ)[35] on the def2-SVP geometries in order to check the basis set and dispersion effects. A fully all-electron def2 basis set was applied to iron. All stationary points were verified using analytic frequencies (no imaginary frequencies for minima, exactly one for transition states). The quintet reactant initially had a residual torsional imaginary frequency of -42 cm^{-1} in the Tp ring; after re-optimization from a geometry displaced by this mode a true minimum was obtained (NImag = 0, the lowest real frequency being $+17.6$ cm^{-1}), and this minimum is 0.23 kcal/mol (electronic) and 1.13 kcal/mol (ΔG_{298}) lower than the previous one, with this shift being uniform and having no effect on the relative ordering or on the margins (the smallest relevant margin being 7.4 kcal/mol, the MECP remaining below the ^{3}β-H TS at def2-SVP). Figure 3 keeps the original reference for consistency with the released logs (SI Section S3.5).

High level (energetics), Fe(I) system: OPBE/def2-SVP for the reference network and the seam-frame single points (no MECP was converged for this system; see Sections 5.3 and 6.2), and OPBE/def2-TZVP single points on

geometries that have been fully DFT-optimized (as stated in Section 5.3); OPBE is employed instead of UB3LYP since a low-Hartree-Fock functional is necessary in order to reveal the two-state crossing in this system (as mentioned in Section 4); TPSSh gives the expected result whereas UB3LYP does not.

Optimization, SCF and thermochemistry settings (as used in Gaussian 16[36]) involve the use of the program's standard convergence criteria for geometry optimizations (a maximum force of 4.5×10^{-4}, an RMS force of 3.0×10^{-4}, a maximum displacement of 1.8×10^{-3} and an RMS displacement of 1.2×10^{-3} au), together with analytic initial Hessians in the case of saddles. Throughout the calculation, we use a pruned (99,590) integration grid and a quadratically convergent SCF fallback. The thermal corrections (ΔG_{298}) are based on rigid-rotor/harmonic-oscillator values derived from the unscaled harmonic frequencies at 298.15 K and 1 atm (these being the program's default settings).

MECP localization: easyMECP[16] (based on Harvey MECP code[14]) at the system’s reference functional/def2-SVP. This was the case for Tp-Fe; for the Fe(I) seam it was not, and the crossing was instead described by the multireference route given below (Sections 5.3, 6.2).

For the ligand-loss scans (M1), relaxed GFN2-xTB calculations are performed for each metal–ligand distance, beginning at the equilibrium value r_0 and extending to r_0 + 2.5 Å in 0.1 Å increments. For η^2 ligands, the scan is conducted along the direction from the metal to the midpoint of the

two bonded atoms. DFT single-point energy calculations on the scan frames and MECP searches utilize the same theoretical level as that used for MECP localization (the reference functional of each system with def2-SVP), with a maximum of 300 MECP steps. To facilitate comparison with ref. 37, the $Fe(CO)_5$ calculations were repeated using B3PW91* (B3PW91 with 15% exact exchange) and def2-TZVP. The minima were confirmed by frequency calculations with the same functional.

Spin–orbit coupling: state-averaged CASSCF using the five Fe 3d orbitals, specifically a CASSCF(6,5) approach for the Tp-Fe(II) (d^6) system and a CASSCF(7,5) one for the Fe(I) (d^7) system, with equal weighting given to the 5 quintet and 10 triplet roots (in the case of Tp-Fe) or to the 5 quartet and 10 doublet roots (for Fe(I)), at the def2-SVP MECP (for Tp-Fe) or seam-frame (for Fe(I)) geometry, and then a QDPT was carried out using the spin–orbit mean-field (SOMF) operator with the def2-SVP/def2-JK and RIJCOSX methods (ORCA 6.1[17]).

Multireference crossing map, Fe(I) system: using SA-CASSCF(7,5)/def2-SVP (pyscf 2.13.0) with an AVAS-selected Fe 3d active space, three roots are taken in each spin block with equal weights of 1/3 and a spin-constrained CI solver, starting from a ROHF reference that has been converged to second order with a level shift of 0.4 au, and then applying a strongly contracted NEVPT2 in which all the orbitals that are doubly occupied and inactive are correlated (the implementation does not include a frozen-core option); the double-shell SA-CASSCF(7,10) check

follows the same protocol but uses an AVAS active space that includes both Fe 3d and 4d orbitals (see SI Sections S4.6 and S6.3).

AI-assisted software development: the code of the extension was developed with the assistance of Claude Code (Anthropic), an AI coding assistant. All generated code was reviewed and tested by the author, and every numerical result reported here was produced by the released code and its archived logs (Data Availability Statement).

4. Level-of-theory dependence of spin gaps and crossing detection

4.1 Semiempirical geometries versus DFT spin-gap energetics

Three lines of evidence establish that for these Fe systems a cheap method can supply spin-resolved *geometries and topology* but not spin-gap *energetics*, motivating the GFN2-explorer / DFT-refiner division of labor:

PM7[18,] as invoked through the stock interface, is spin-blind and additionally inverts the Fe spin ordering: our direct diagnostic on the Tp-Fe(II)-Et reactant yields the result singlet < triplet < quintet, which corresponds to a low-spin ground state for a complex that is experimentally known to be high-spin (quintet) (SI Section S7.1).

A foundational MLP misorders the system's spin states. When MACE-OMol25[38,39] was tested on two systems using high-level reference data—$Fe(CO)_4$ (singlet−triplet, CCSD(T)[40]) and $[Fe(H_2O)_6]^{2+}$ (singlet−quintet, CASPT2[41])—it made an error of −12.0 kcal/mol in the first case, large enough to reverse the sign of the gap and place the ground state on the

incorrect surface, and an error of −24.4 kcal/mol in the second case, giving an RMSE of 19.2 kcal/mol for the two systems together (SI Section S7.2).

GFN2-xTB[19] handles spin correctly and produces separate spin surfaces, but it still predicts the wrong ground-state spin for Tp-Fe(II)-Et (low-spin), so its spin-gap energetics are not quantitative (SI Section S7.1). In this context, it is used solely for its reliable features, specifically the geometries and connectivity on each spin surface. Its gap errors are accommodated by the large M1 window (SI Section S3.1), and no GFN2-xTB energy is incorporated into crossing energies or rates. MACE-OMol25 was assessed exclusively as a DFT substitute for these energetics, not as the exploration engine.

The workflow uses GFN2-xTB to *identify* crossings and provide geometries, and employs DFT (or, in principle, a more highly correlated method) for all spin-sensitive energetics.

4.2 Functional dependence of crossing detection

The choice of DFT functional for detecting spin crossings is not neutral. In the case of the Fe(I) isomerization system (Section 5.3), UB3LYP (which contains 20% Hartree-Fock exchange) fails to show the two-state crossing because it stabilizes the higher-spin surface too much and thus keeps the doublet channel from being a contender over the relevant geometries; on the other hand, the low-HF functionals OPBE and TPSSh both show the crossing as well as the spin inversion along the oxidative-addition coordinate. Single-determinant spin-gap energetics depend strongly on the

amount of exact exchange,[42,43] so, for automated *detection* of a spin crossing, a low-HF functional should be the default choice. We therefore recommend using OPBE (or a similar low-HF GGA) as the default functional, with a hybrid employed only as a check. A third finding relating to the level of theory concerns the geometry level: a DFT//semiempirical single-point ladder is capable of completely reversing a near-degenerate two-state ordering, as the Fe(I) system shows in Section 5.3, and near-degenerate spin-sensitive saddles must therefore be optimized at a consistent DFT level.

5. Results and validation

5.1 Validation on simple cases

The ability of the LL engine to handle spin was checked on H_2CO and on $Fe(CO)_5$, in the case of which the singlet and triplet calculations yield genuinely different LL surfaces (with 3 versus 6 transition states; each saddle point having one imaginary frequency) and GFN2, in contrast to PM7, correctly treats $Fe(CO)_5$ as a singlet. The Sella TS/IRC layer reproduces the H_2CO network using a list of occupied TSs and IRC-verified edges. These two systems demonstrate blind, spin-resolved network discovery (Section 2.8).

The reaction pathway for $Fe(CO)_5$ features a spin crossing along the CO-loss coordinate, represented by the process $^1Fe(CO)_5 \rightarrow {}^3Fe(CO)_4 + CO$, which is not captured by the vertical gaps at network nodes. The ligand-loss scan (Section 2.4) flags both CO-loss channels. The DFT curves intersect only in

the axial channel, and M2 converged from this seed without manual intervention. The MECP exhibits C_s symmetry and, at B3PW91*/def2-TZVP, displays the incoming CO at a distance of 2.32 Å and an Fe–C–O angle of 135°, consistent with the structure reported by Harvey and Aschi (2.24 Å, 135°).[37] The MECP is located 1.6 kcal/mol (B3PW91*/def2-TZVP) and 1.4 kcal/mol (UB3LYP/def2-SVP) above $^3Fe(CO)_4$ + CO in electronic energy. Their zero-point-corrected values range from 0.5 to 3.7 kcal/mol, depending on the functional (0.49 kcal/mol with B3PW91*). In the graph (M3), the MECP connects the singlet minimum of $Fe(CO)_5$ to the triplet $Fe(CO)_4$ + CO channel. A rate constant was not calculated because spin–orbit coupling was not assessed. Additional details, including the singlet instability at the MECP and the triplet network channels, are provided in SI Section S8.

5.2 Tp-Fe(II)-ethyl β-hydride elimination

Spin-crossing handling for the β-hydride saddles. For this system, the spin crossing is facile and not rate-determining, as it enables access to the lower triplet TS, which governs the reaction rate. The following subsection explains how we reached this conclusion without operator intervention after locating the saddles (Figure 3).

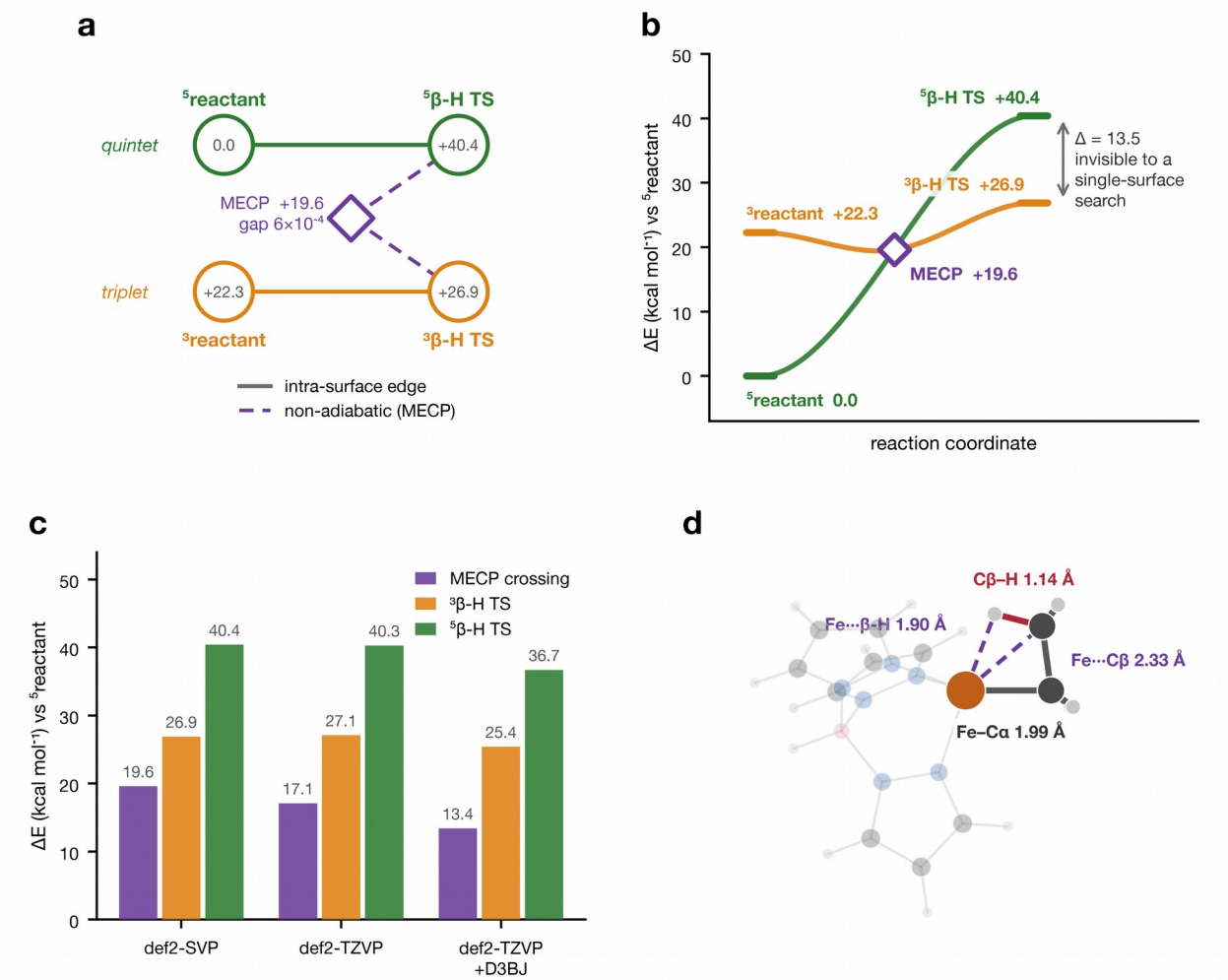


Figure 3. Two-state reactivity of Tp-Fe(II)-ethyl β-hydride elimination. (a) Automatically constructed two-state reaction network (M3 output) including, on both the quintet (green) and triplet (orange) surfaces, a reactant and a β-hydride elimination TS; the $^{5}/^{3}$ MECP (open purple diamond, +19.6 kcal/mol at def2-SVP, inter-surface gap 6×10^{-4} kcal/mol) connects the two surfaces through nonadiabatic edges (dashed purple). (b) Reaction energy profile showing the more favorable triplet TS. (c) The order MECP < ^{3}β-H TS < ^{5}β-H TS is unaffected by changes in basis set or dispersion. (d) MECP geometry (UB3LYP/def2-TZVP) corresponds to the

agostic approach that occurs before β-hydride transfer. The Tp ligand has been faded for clarity.

The two per-spin β-hydride saddles in the Tp-Fe(II)-Et complex were found using targeted Sella saddle optimization, the optimization being initiated by a relaxed Cβ–H scan (Section 2.8). Once these saddles were identified, the rest of the chain was automatically generated. The M0 check shows that all four Tp-Fe stationary states exhibit single-reference character when calculated using UB3LYP, OPBE and TPSSh (SI Section S2.4). The spin-state enumeration (M1) identifies the quintet β-H region as a potential quintet↔triplet close-lying MECP. Automatic localization of the MECP (M2) from the identified geometry leads to the location of a quintet/triplet MECP with an inter-surface gap of 6×10^{-4} kcal/mol (indicating genuine degeneracy) at +19.6 kcal/mol with respect to the quintet reactant ground state (def2-SVP). The geometry of the MECP lies on the agostic approach that comes before the hydrogen transfer, showing an Fe···Cβ distance of 2.36 Å (2.66 Å at the triplet transition state) and the migrating β-hydrogen still bonded to Cβ at 1.14 Å while agostically interacting with Fe through a Fe···H distance of 1.95 Å. When the same crossing is re-localized at def2-TZVP (Figure 3d, SI Section S3.6), both the degeneracy (with a gap of 1×10^{-3} kcal/mol) and the geometry are retained to within 0.06 Å at +16.7 kcal/mol. When included in the graph (M3), the MECP connects the quintet and triplet β-H transition-state regions through nonadiabatic crossing edges (Figure 3a).

Energetics and nonadiabatic rate. The reaction rate is governed by the triplet β-hydride transition state, and the associated spin crossing is not a kinetic bottleneck. The supporting evidence is presented below.

At the UB3LYP/def2-SVP level, the energy ladder with respect to the ground state of the quintet reactant (as shown in Figure 3b) is: triplet reactant +22.3, triplet β-H TS +26.9, quintet β-H TS +40.4, MECP crossing +19.6 kcal/mol, which agrees well with the previous manual investigation of this system[44] (triplet reactant +22.5, ^{3}β-H TS +25.2 kcal/mol). When the values are compared directly, our own ΔG_{298} for the triplet β-H TS is +27.9 kcal/mol, which is 2.7 kcal/mol higher than that value. The easy crossing confirmed in this study applies to this truncated ligand set only: the steric effects of the substituents which make the experimental Tp complexes resistant to β-hydride elimination (with a barrier of 31.8 kcal/mol in the same study for TpiPr,iPr) are not included in the model. Two-state β-hydride elimination in Fe(II)/Co(II) alkyls has since been re-examined independently by means of an automated reaction-path search, with the explicit characterization of the spin-inversion points.[45]

The value of the spin–orbit coupling at the MECP, as calculated by CASSCF/ QDPT-SOMF, is $H_{SO} \approx 279$ cm^{-1} (and this is within the normal range for Fe(II)). The NA-TST rate for the crossing is high ($\kappa \approx 0.77$; k(298 K) ≈ 1.5×10^{-3} s^{-1} for the hop itself), being about four orders of magnitude faster than the chemical turnover. The microkinetic analysis finds that the spin-cross route (with an effective barrier of 26.9 kcal/mol and the triplet TS

being the rate-determining step) is the one that is operative rather than the same-spin quintet route (40.4 kcal/mol; on the ΔG_{298} scale the TS-TS energy difference decreases from 13.5 to 9.8 kcal/mol, yet the same conclusion is reached): the spin crossing is easy and not rate-determining because it provides access to the lower triplet transition state, which is rate-controlling. This two-state-reactivity model is obtained when the M1–M4 layer runs unsupervised using the two saddles located, together with a rate assigned to the crossing.

5.3 Fe(I)-catalyzed alkene isomerization: two-state oxidative addition

The second example involves a two-state catalytic cycle: in the case of alkene isomerization catalyzed by an iron(I) complex containing a bis(carbene)borate group, its two-state reactivity has been suggested to confer resistance to σ-donor inhibitors.[25] The proposed mechanism involves the substrate binding on the quartet (S = 3/2) surface, whereas the oxidative addition to an η^1-allyl intermediate occurs only on the doublet (S = 1/2) surface, a change in spin that can only take place after the alkene has bound. We have modeled a truncated but accurate active complex (comprising 38 atoms; neutral, obtained by combining the bookkeeping monoanionic bis(carbene)borate L^- with Fe(I) d^7 and a neutral alkene to give an overall charge of zero). The M0 check is passed with UB3LYP, OPBE and TPSSh (the quartet $\langle S^2 \rangle$ values being 3.751–3.755 and the doublet values 0.750–0.751 after annihilation, with the annihilation causing a

lowering; see SI Section S2.2), unlike the nitrene (Section 5.4). In agreement with Section 4.2, the two-state crossing is revealed by OPBE and TPSSh but not by UB3LYP; OPBE/def2-SVP is thus taken as the reference functional for this system, with TPSSh used as a cross-check.

Two-state network and the $^4/^2$ seam (Figure 4a). M1 flags the binding→OA region as a quartet↔doublet crossing candidate; the doublet oxidative addition transition state is found by means of targeted saddle point optimization (as described in Section 2.8) and is verified using an IRC (with a single imaginary frequency, $\nu = -541$ cm^{-1}); a nearly degenerate $^4/^2$ seam frame is located with an inter-surface gap of approximately 0.01 kcal/mol, even though, as shown below, it is not an energy-minimized MECP; the spin–orbit coupling at that frame, calculated by CASSCF/QDPT-SOMF, is $H_{SO} \approx 90$ cm^{-1} (RMS), which is consistent with a first-row two-state crossing and is smaller than the Tp-Fe value of 279 cm^{-1} as is expected given the different orbital character of the crossing.

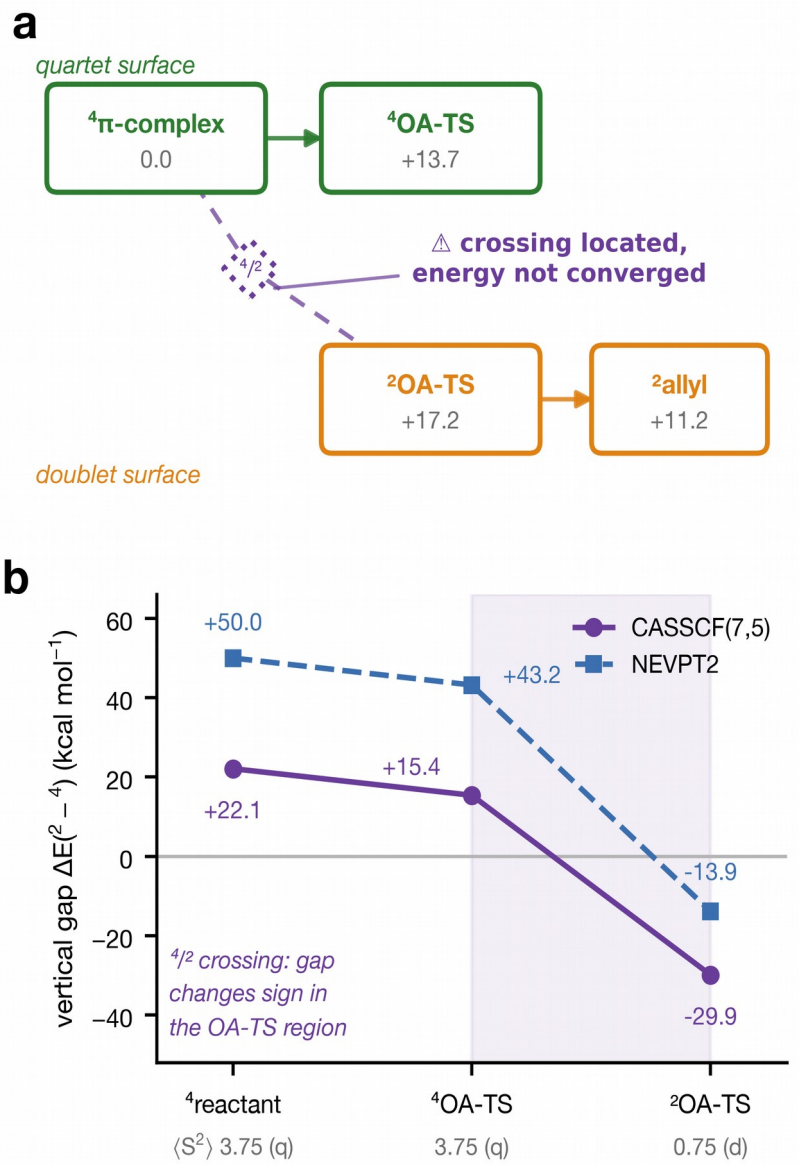


Figure 4. Vertical gaps and the unconverged crossing for the Fe(I) two-state network. (a) Quartet lane: ^{4}π-complex (0.0) → ^{4}OA-TS (+13.7); doublet lane: ^{2}OA-TS (+17.2) → ^{2}allyl (+11.2). The $^{4}/^{2}$ crossing is located but its energy is not converged (dotted diamond): near-degeneracy prevents convergence of the single-determinant SCF, and the crossing was handled by the manually configured multireference fallback. (b) Vertical gaps $\Delta E(^{2}-^{4})$ from SA-CASSCF(7,5) (purple) and SC-NEVPT2 (blue) at the three DFT-optimized geometries; the sign change between the ^{4}OA-TS and ^{2}OA-TS

geometries (shaded) places the $^4/^2$ crossing in the OA-TS region. UDFT $\langle S^2 \rangle$ values after annihilation, at each geometry's own optimized state, are given below the axis.

Energetics: consistent-level ladder. For a DFT//GFN2 calculation (DFT single points on GFN2 geometries) the doublet OA-TS appeared ~12.5 kcal/mol *below* the quartet OA-TS, implying an effective OA barrier of only ~+3.9 kcal/mol and a strongly preferred doublet channel. When all four stationary points are re-optimized at a consistent DFT level this result does not survive: the two OA-TSs are near-degenerate, with the quartet marginally lower, for both basis sets. Figure 5 shows the two levels of theory side by side.

The gap between the doublet and quartet OA-TS states is +2.65 kcal/mol when using def2-SVP, +3.55 kcal/mol with def2-TZVP, and +3.37 kcal/mol when D3(BJ) dispersion is included (the quartet remains lower in all cases; the ΔΔE values are +0.47 and +1.37 kcal/mol at the two bases, see SI Section S4.3) — the states being nearly degenerate at all levels. Increasing the basis set to TZVP does not recover the //GFN2 trend, and the D3(BJ) term (approximately −19 kcal/mol) is almost the same for all four species, causing a shift in the ladder of at most 0.3 kcal/mol. The spin ground state still inverts as we move along the oxidative-addition coordinate — the quartet π-complex reactant (0.0) transforms into a doublet η^1-allyl product (+11.2) — so a $^4/^2$ crossing must occur. The turnover oxidative-addition barrier is about 14 to 17 kcal/mol (the quartet ^{4}OA-TS at a ΔG of +13.7 kcal/

mol represents the lower saddle), and the two spin pathways are competitive at the OA-TS: the productive doublet η^1-allyl product is obtained through the crossing, not as a result of a large kinetic preference for a doublet-TS.

Role of the substrate **(σ-base resistance).** Since the catalyst alone has a large quartet→doublet energy gap (approximately +25.6 kcal/mol at OPBE/def2-SVP; see SI Section S4.4), the spin change cannot take place in the absence of a substrate; however, when the alkene is bound the two spin surfaces become nearly degenerate at the crossing seam (the gap being approximately 0.01 kcal/mol) and the desired doublet product is obtained along the oxidative-addition pathway (Figure 6). A σ-donor that binds in place of the substrate is unable to produce this near-degenerate crossing region, which is the structural feature underlying the observed σ-base resistance.[25] The reasoning is based on this general contrast between the catalyst alone and when it is bound to the alkene, not on the absolute energy of the crossing point when the alkene is bound, which has not been converged (below).

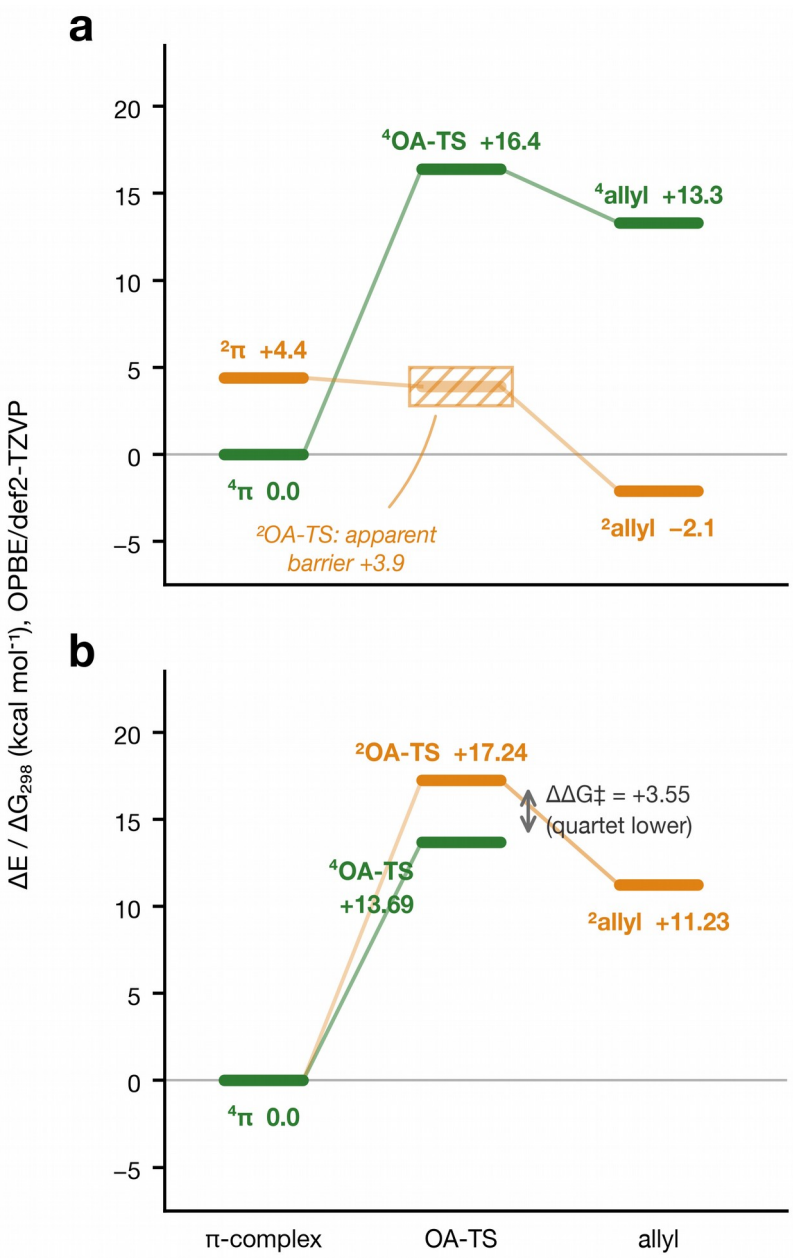


Figure 5. A mixed-level ladder can invert a near-degenerate two-state ordering. (a) OPBE/def2-TZVP single points on GFN2-xTB geometries place the doublet OA-TS about 12.5 kcal/mol below the quartet saddle, giving an apparent doublet barrier of +3.9 kcal/mol. (b) At consistent-level OPBE/def2-TZVP//OPBE/def2-SVP ΔG_{298} the ordering inverts: the saddles

are nearly degenerate, the quartet lower by $\Delta\Delta G^{\ddagger}$ = +3.55 kcal/mol (def2-TZVP; +3.37 with D3(BJ)).

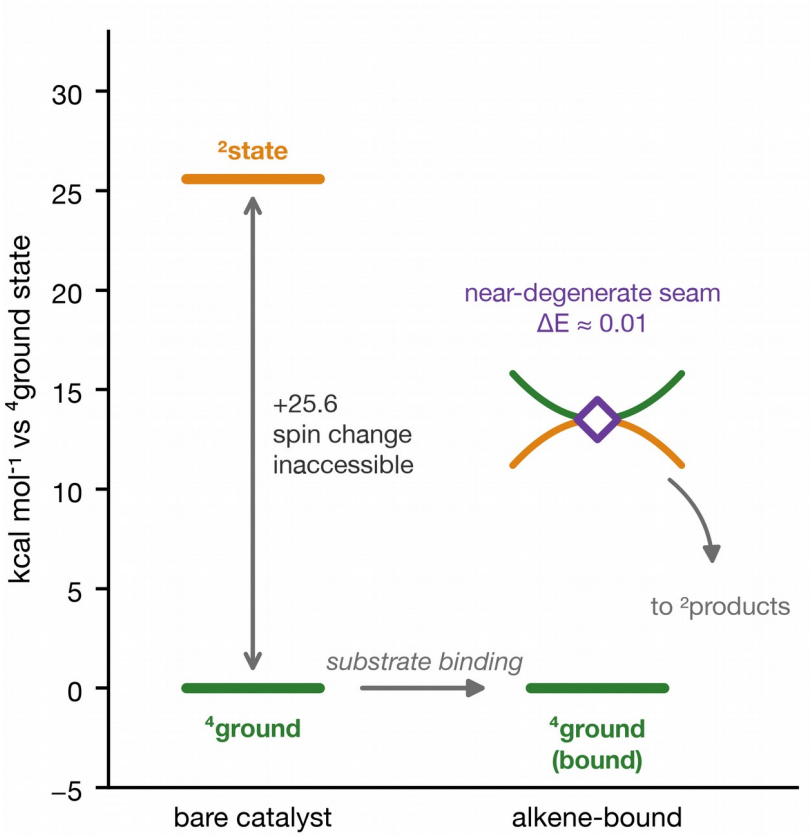


Figure 6. Substrate binding enables the spin change (σ-base resistance). In the bare Fe(I) catalyst the doublet lies +25.6 kcal/mol (OPBE/def2-SVP) above the quartet ground state; with the alkene bound, the two surfaces are nearly degenerate at the crossing seam (gap ≈0.01 kcal/mol), opening the spin-change pathway along the oxidative-addition coordinate toward the doublet product. The seam is placed illustratively on the energy axis.

The crossing energy. The seam-crossing frame located by the extension in this instance is not a true MECP, since the one seam energy that could be obtained with a single determinant (the quartet single point at OPBE/def2-SVP) is found 27 kcal/mol above the quartet reactant — *that is, above* both

of the oxidative-addition transition states, a situation which is impossible for a real MECP; consequently the energy of the frame cannot be used as $dE_{crossing}$. Since the single-determinant doublet SCF at the seam is numerically ill-conditioned (Section 6.2), the crossing was instead characterized using a balanced multireference approach — SA-CASSCF(7,5) with SC-NEVPT2, with both spin states treated on an equal basis at the DFT-optimized stationary points — a method which serves as a spin-gap / crossing-location diagnostic and not as a reevaluation of the barrier (Section 6.2). The vertical doublet-minus-quartet gap $\Delta E(^2-^4)$ is +22.1 (CASSCF) / +50.0 (NEVPT2) kcal/mol at the ^{4}reactant π-complex, +15.4 / +43.2 at the quartet-optimized ^{4}OA-TS, and −29.9 / −13.9 at the doublet-optimized ^{2}OA-TS (Figure 4b): the gap changes sign in the oxidative-addition transition-state region at both levels, so the spin flip occurs simultaneously with oxidative addition, in agreement with the independent SOC-CASSCF calculation at the stored seam (the doublet being 22.4 kcal/mol higher than the quartet, i.e., on the reactant side). The *location* of the crossing is thus established, although its kinetic weight is not. A NA-TST hop rate requires the crossing energy on the DFT reaction-energy scale, a quantity which the minimal active space does not provide, and SI Section S4.7 (Figure S2) demonstrates that the *sign* of the kinetic verdict depends on that energy: at the archived seam parameterization the hop is 1.7 orders of magnitude faster than oxidative-addition turnover when using the single archived $dE_{crossing}$ of 9.0 kcal/mol, but 1.7 to 4.3 orders of magnitude slower

everywhere within the +13.7 to +17.2 kcal/mol OA-TS window where the multireference mapping places the seam. No rate is therefore given for this system; the determination of a converged crossing energy on the DFT scale is the one outstanding computational refinement (Section 6.2).

5.4 ⟨S²⟩ check for an iron-nitrene C–H amination

A third system, intermolecular C–H amination using an iron-nitrene, namely Fe(porphine)(–OCH_3)(=N–SO_2CH_3) together with toluene, involving benzylic hydrogen atom transfer, provided a case study where the M0 check identifies a genuinely multireference system. Iron catalysts are capable of carrying out intermolecular amination of benzylic C(sp^3)–H bonds;[46] the iron-porphyrin nitrene has been examined theoretically as the active species in the case of intermolecular C–H amination for a closely related model — that of an unsubstituted porphyrin iron nitrene with an axial methoxide, in which instead of a methanesulfonyl group there is a tosyl group and the nitrene abstracts a hydrogen atom from *p*-ethylanisole rather than from toluene — the process taking place by means of hydrogen-atom abstraction followed by radical recombination;[47] and under catalytic conditions with a bis(pyrrolyl)pyridine iron catalyst a large primary kinetic isotope effect ($k_H/k_D \approx 61$ for xanthene-d_2) indicates that the C–H bond cleavage step is rate-determining.[48] The extension therefore has to deal with the hydrogen-atom-transfer transition state.

Because blind reactive sampling could not access the bimolecular Fe=N···H–C hydrogen-atom-transfer geometry (Section 6.1), we located and

characterized the two HAT transition-state geometries directly. The triplet HAT transition state has a value of $\langle S^2 \rangle$ much higher than the ideal value of 2.0 (under UB3LYP, the Stable=Opt-confirmed SCF solution: 4.22 → 5.60; under OPBE: 3.97 → 4.50; under TPSSh: 4.25 → 5.68), the spin annihilation *effect increasing* rather than decreasing this value — this is the diagnostic sign of a truly multiconfigurational state, and this behavior is independent of the functional. The quintet HAT transition state itself passes the check (the spin annihilation brings $\langle S^2 \rangle$ down to within 0.4 of the ideal 6.00 for all three functionals), and the clean quintet HAT barrier is robust (+39.2 UB3LYP, +45.1 OPBE, +39.2 TPSSh kcal/mol; electronic energies at functional/def2-SVP relative to the reactant complex, source energies given in SI Section S2.3), but the triplet surface which would be involved in any two-state pathway is not representable by a single-reference approach. The M0 check marks this node as being out of scope.

6. Discussion

The three systems discussed above illustrate the different possible outcomes of the extension (as shown in Figure 1): a converged MECP and rate for Tp-Fe, a located crossing with an unconverged energy for Fe(I), and a multireference flag for the iron-nitrene.

6.1 Scope of blind reactive sampling

For the nitrene system, the limit of blind BXDE/vdW sampling was also determined: beginning with a non-covalent Fe-nitrene···substrate complex, blind reactive sampling is unable to get to the bimolecular association/HAT

geometry: the substrate dissociates prior to a reactive encounter being sampled, on all the spin surfaces and at all the model rungs that have been examined. Although blind sampling is able to handle unimolecular rearrangement on the small systems investigated above (H_2CO, $Fe(CO)_5$), it cannot achieve the intermolecular association step in a bimolecular reaction, which means *that targeted* exploration (such as constrained sampling or seeding from a pre-formed encounter complex) is required. The second limitation is due to size rather than to molecularity: on the 34-atom Tp complex blind BXDE is limited by decomposition and does not reach the *unimolecular* β-H product either (Section 2.8). These two limits constrain the sampler from opposite directions.

6.2 SCF convergence at near-degenerate crossing seams

It is not possible, using a single-determinant method, to determine the Fe(I) $^4/^2$ crossing energy. Two difficulties arise. The first is that the SCF along the crossing path is ill-conditioned: on the interpolated-path seam seed the DIIS method diverges (the energy fluctuates by hartrees in the wrong direction), an aggressively level-shifted DIIS converges to a false, high-lying determinant, and a full quadratic SCF with level shift and stability search is stable but needs hours of coupled-perturbed iterations without taking any macro-steps (the failure details are given in SI Section S6.2). The second, and more fundamental, issue is that the *geometry* of the seam itself is important: when a linearly interpolated seed is used both the doublet and quartet SCF collapse to the same spurious determinant (~900 kcal/mol

above the reactants), whereas if a geometry that has been physically relaxed along the SOC-seam trajectory is used the quartet converges properly (after 27 cycles). The multiconfigurational nature of the doublet, the fragility of the SCF algorithm, and the unphysical character of the trial geometry all play a part in this, and separating out these factors would require a converged MECP optimization that could not be carried out. Two guidelines therefore apply in automated MECP work: (i) when the two surfaces are nearly degenerate over an extended region of the seam, start the crossing search from geometries that have been relaxed — for surfaces that are well separated and have a local crossing a linear interpolant can still be used as an adequate and inexpensive seed (the Tp-Fe MECP in Section 5.2 converged from a 60/40 reactant/TS interpolant; the reactant-side energy gap is 22 kcal/mol), and (ii) if single-determinant SCFs on both spin states behave badly at the same point, fall back on a multireference or state-averaged approach (CASSCF/NEVPT2). The present case was solved in exactly this way (Section 5.3, Figure 4b); to obtain the crossing energy on the DFT reaction-energy scale either a robust seam-SCF procedure or a larger active space with matched geometry optimization would be needed. Two approaches could yield a converged crossing energy at such seams: MECP optimization on multireference surfaces, where the MECP algorithm requires only the energies and gradients of the two states (for example, SA-CASSCF),[14,49] and SCF procedures that converge to a specified state, such

as the maximum-overlap method.[50] Neither approach is implemented in the present extension.

The minimal active space that is being considered is defined by four points. The first of these is the (7,5) space, which corresponds to the Fe 3d shell as selected by AVAS, with three roots assigned on a state-averaged basis per spin block; this is the naturally minimal option for a d^7 system, and it deliberately excludes the Fe 4d double shell as well as the σ/σ* orbitals associated with the reorganizing C–H/Fe–C bonds, which is precisely the reason why it manages to reproduce the vertical spin *gaps* but not the reaction *barriers* (its quartet "barrier" at the ^{4}OA-TS is +49 kcal/mol compared to the DFT value of +13.7 kcal/mol). The second point is that $\Delta E(^2-^4)$, as determined by the location argument and compared between two geometries treated in the same way, changes sign at both the CASSCF and NEVPT2 levels independently; the dynamic-correlation shift is large and depends on the geometry (+28 kcal/mol at the reactant and at the ^{4}OA-TS, +16 at the ^{2}OA-TS — NEVPT2 consistently disfavors the doublet) without causing the crossing to move outside the OA-TS window. The third point is that a similar conclusion is obtained from the independent SOC-CASSCF calculation carried out at the stored seam frame (the doublet is above the quartet on the reactant side). The fourth point is that a double-shell SA-CASSCF(7,10) (including the AVAS Fe 3d and 4d shells, with three roots per spin) at the same two OA-TS geometries also retains the sign change (the value of $\Delta E(^2-^4)$ is +42.6 kcal/mol at the ^{4}OA-TS geometry and −4.7

kcal/mol at the ^{2}OA-TS geometry (def2-SVP)). The inclusion of the double shell stabilizes the quartet relative to the (7,5) values by a comparable amount at both geometries (+27.2 and +25.2 kcal/mol), leading to a substantial increase in the magnitude of the gap while keeping the crossing within the OA-TS window, so the conclusion regarding the location is stable with respect to the active space at the variational level. This is the only level at which the double-shell result is presented: when SC-NEVPT2 is applied to the double-shell state-averaged space, stable perturbation energies are not obtained in this case because the perturbation-step CASCI is inconsistent with the converged state-averaged solution (see SI Section S4.6), and the (7,5) SC-NEVPT2 result from Figure 4b serves as the correlated check. The CASSCF/NEVPT2 protocol is provided in SI Section S6.3.

7. Conclusion

Automated exploration of reaction networks has been extended to include spin resolution, with an explicit check for multireference character. Application to three iron systems yielded the three possible outcomes. For Tp-Fe(II)-ethyl β-H elimination, the spin-resolved layer identified the quintet/triplet crossing, converged the MECP at two basis sets (def2-SVP +19.6, def2-TZVP re-localized +16.7 kcal/mol), assigned an NA-TST rate, and concluded that the crossing is not rate-limiting, with the triplet transition state controlling the rate. This result is consistent with previous manual studies and is robust with respect to basis set and dispersion. For the Fe(I) isomerization catalyst, the extension located the crossing but not

its energy: the $\langle S^2 \rangle$ check was passed, the near-degenerate seam could not be resolved by single-determinant SCF methods, and the CASSCF/NEVPT2 fallback identified the crossing concurrent with oxidative addition, but the absolute energy and hop rate remained unconverged. In the case of iron-nitrene amination, the extension flagged the system as multireference: spin annihilation caused $\langle S^2 \rangle$ to increase across three functionals, indicating a state that cannot be represented by a single determinant.

The availability of this methodology as open-source software enables spin-crossing detection, MECP localization, and rate analysis for 3d transition-metal mechanisms, with clear indications of methodological limitations. The degree of automation differs across stages. Blind path sampling was employed for H_2CO and $Fe(CO)_5$, whereas targeted optimization was used to obtain transition states for the larger iron complexes. Once these stationary points were identified, spin-state enumeration, crossing detection, MECP localization and rate assignment were performed without further manual intervention, except at the Fe(I) seam. Future work should focus on extending blind spin-resolved *discovery* to larger complexes and applying the extension to mechanisms that remain unresolved. Improved accuracy of spin gaps at the exploration level would permit a narrower M1 window. Relevant methods include spin-polarized tight-binding approaches[51,52] and machine-learned potentials that incorporate both charge and spin as inputs,[53] once fine-tuned on spin-resolved DFT data.

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CONFLICT OF INTEREST STATEMENT

The author declares no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available. The spin-resolved AutoMeKin extension (low-level engine, TS/IRC layer, M0 $\langle S^2 \rangle$ check and the M1–M4 modules) together with the analysis and validation scripts is archived at Zenodo (version 1.1.0), DOI 10.5281/zenodo.23370373. The AutoMeKin extension is developed at https://github.com/joshazerty/AutoMeKin. The validation cases (H_2CO, $Fe(CO)_5$, Tp-Fe(II)-ethyl, Fe(I) isomerization) and the nitrene multireference test case are included as worked examples. All geometries, input decks, logs, the M0 $\langle S^2 \rangle$ tables and the rate and microkinetic inputs are contained in that archive and in the Supporting Information of this article.

Keywords: automated reaction discovery, minimum-energy crossing point, multireference diagnostics, spin-forbidden reactivity, transition-metal catalysis

Additional Supporting Information may be found in the online version of this article.

References and Notes

1. E. Martínez-Núñez, G. L. Barnes, D. R. Glowacki, et al., “AutoMeKin2021: An open-source program for automated reaction discovery,” Journal of Computational Chemistry 42 (2021): 2036, 10.1002/jcc.26734.

2. E. Martínez-Núñez, G. L. Barnes, C. Bo, et al., AutoMeKin, version rev1137 (Zenodo, 2024), 10.5281/zenodo.12564112.

3. J. P. Unsleber, S. A. Grimmel, and M. Reiher, “Chemoton 2.0: Autonomous Exploration of Chemical Reaction Networks,” Journal of Chemical Theory and Computation 18 (2022): 5393, 10.1021/acs.jctc.2c00193.

4. S. Maeda, K. Ohno, and K. Morokuma, “Systematic exploration of the mechanism of chemical reactions: the global reaction route mapping (GRRM) strategy using the ADDF and AFIR methods,” Physical Chemistry Chemical Physics 15 (2013): 3683, 10.1039/c3cp44063j.

5. A. Hashemi, S. Bougueroua, M.-P. Gaigeot, and E. A. Pidko, “ReNeGate: A Reaction Network Graph-Theoretical Tool for Automated Mechanistic Studies in Computational Homogeneous Catalysis,” Journal of Chemical Theory and Computation 18 (2022): 7470, 10.1021/acs.jctc.2c00404.

6. R. Van de Vijver and J. Zádor, “KinBot: Automated stationary point search on potential energy surfaces,” Computer Physics Communications 248 (2020): 106947, 10.1016/j.cpc.2019.106947.

7. J. N. Harvey, “Understanding the kinetics of spin-forbidden chemical reactions,” Physical Chemistry Chemical Physics 9 (2007): 331, 10.1039/b614390c.

8. D. Schröder, S. Shaik, and H. Schwarz, “Two-State Reactivity as a New Concept in Organometallic Chemistry,” Accounts of Chemical Research 33 (2000): 139, 10.1021/ar990028j.

9. S. Shaik, H. Hirao, and D. Kumar, “Reactivity of High-Valent Iron–Oxo Species in Enzymes and Synthetic Reagents: A Tale of Many States,” Accounts of Chemical Research 40 (2007): 532, 10.1021/ar600042c.

10. Y. Harabuchi, T. Taketsugu, and S. Maeda, “Combined gradient projection/single component artificial force induced reaction (GP/SC-AFIR) method for an efficient search of minimum energy conical intersection (MECI) geometries,” Chemical Physics Letters 674 (2017): 141, 10.1016/j.cplett.2017.02.069.

11. K. Saita, M. Takagi, Y. Harabuchi, H. Okada, and S. Maeda, “Exploring potential crossing seams in periodic systems: Intersystem crossing pathways in the benzene crystal,” Journal of Chemical Physics 149 (2018): 072329, 10.1063/1.5025023.

12. J. P. Janet and H. J. Kulik, “Predicting electronic structure properties of transition metal complexes with neural networks,” Chemical Science 8 (2017): 5137, 10.1039/c7sc01247k.

13. A. Nandy, C. Duan, M. G. Taylor, F. Liu, A. H. Steeves, and H. J. Kulik, “Computational Discovery of Transition-metal Complexes: From High-

throughput Screening to Machine Learning,” Chemical Reviews 121 (2021): 9927, 10.1021/acs.chemrev.1c00347.

14. J. N. Harvey, M. Aschi, H. Schwarz, and W. Koch, “The singlet and triplet states of phenyl cation. A hybrid approach for locating minimum energy crossing points between non-interacting potential energy surfaces,” Theoretical Chemistry Accounts 99 (1998): 95, 10.1007/s002140050309.

15. A. O. Lykhin, D. S. Kaliakin, G. E. dePolo, A. A. Kuzubov, and S. A. Varganov, “Nonadiabatic transition state theory: Application to intersystem crossings in the active sites of metal-sulfur proteins,” International Journal of Quantum Chemistry 116 (2016): 750, 10.1002/qua.25124.

16. J. Rodríguez-Guerra, easyMECP, version 0.3.2 (Zenodo, 2020), 10.5281/zenodo.4293421.

17. F. Neese, “Software Update: The ORCA Program System—Version 6.0,” WIREs Computational Molecular Science 15 (2025): e70019, 10.1002/wcms.70019.

18. J. J. P. Stewart, “Optimization of parameters for semiempirical methods VI: more modifications to the NDDO approximations and re-optimization of parameters,” Journal of Molecular Modeling 19 (2013): 1, 10.1007/s00894-012-1667-x.

19. C. Bannwarth, S. Ehlert, and S. Grimme, “GFN2-xTB─An Accurate and Broadly Parametrized Self-Consistent Tight-Binding Quantum Chemical Method with Multipole Electrostatics and Density-Dependent Dispersion

Contributions,” Journal of Chemical Theory and Computation 15 (2019): 1652, 10.1021/acs.jctc.8b01176.

20. A. Hjorth Larsen, J. Jørgen Mortensen, J. Blomqvist, et al., “The atomic simulation environment—a Python library for working with atoms,” Journal of Physics: Condensed Matter 29 (2017): 273002, 10.1088/1361-648x/aa680e.

21. E. D. Hermes, K. Sargsyan, H. N. Najm, and J. Zádor, “Sella, an Open-Source Automation-Friendly Molecular Saddle Point Optimizer,” Journal of Chemical Theory and Computation 18 (2022): 6974, 10.1021/acs.jctc.2c00395.

22. W. Kabsch, “A solution for the best rotation to relate two sets of vectors,” Acta Crystallographica Section A 32 (1976): 922, 10.1107/s0567739476001873.

23. W. Kabsch, “A discussion of the solution for the best rotation to relate two sets of vectors,” Acta Crystallographica Section A 34 (1978): 827, 10.1107/s0567739478001680.

24. S. Kozuch and S. Shaik, “How to Conceptualize Catalytic Cycles? The Energetic Span Model,” Accounts of Chemical Research 44 (2011): 101, 10.1021/ar1000956.

25. S. A. Lutz, A. K. Hickey, Y. Gao, C.-H. Chen, and J. M. Smith, “Two-State Reactivity in Iron-Catalyzed Alkene Isomerization Confers σ-Base Resistance,” Journal of the American Chemical Society 142 (2020): 15527, 10.1021/jacs.0c07300.

26. A. D. Becke, “Density-functional thermochemistry. III. The role of exact exchange,” Journal of Chemical Physics 98 (1993): 5648, 10.1063/1.464913.

27. C. Lee, W. Yang, and R. G. Parr, “Development of the Colle-Salvetti correlation-energy formula into a functional of the electron density,” Physical Review B 37 (1988): 785, 10.1103/physrevb.37.785.

28. P. J. Stephens, F. J. Devlin, C. F. Chabalowski, and M. J. Frisch, “Ab Initio Calculation of Vibrational Absorption and Circular Dichroism Spectra Using Density Functional Force Fields,” Journal of Physical Chemistry 98 (1994): 11623, 10.1021/j100096a001.

29. N. C. Handy and A. J. Cohen, “Left-right correlation energy,” Molecular Physics 99 (2001): 403, 10.1080/00268970010018431.

30. J. P. Perdew, K. Burke, and M. Ernzerhof, “Generalized Gradient Approximation Made Simple,” Physical Review Letters 77 (1996): 3865, 10.1103/physrevlett.77.3865.

31. M. Swart, A. W. Ehlers, and K. Lammertsma, “Performance of the OPBE exchange-correlation functional,” Molecular Physics 102 (2004): 2467, 10.1080/00268970420002750l7.

32. J. Tao, J. P. Perdew, V. N. Staroverov, and G. E. Scuseria, “Climbing the Density Functional Ladder: Nonempirical Meta–Generalized Gradient Approximation Designed for Molecules and Solids,” Physical Review Letters 91 (2003): 146401, 10.1103/physrevlett.91.146401.

33. V. N. Staroverov, G. E. Scuseria, J. Tao, and J. P. Perdew, “Comparative assessment of a new nonempirical density functional: Molecules and

hydrogen-bonded complexes,” Journal of Chemical Physics 119 (2003): 12129, 10.1063/1.1626543.

34. F. Weigend and R. Ahlrichs, “Balanced basis sets of split valence, triple zeta valence and quadruple zeta valence quality for H to Rn: Design and assessment of accuracy,” Physical Chemistry Chemical Physics 7 (2005): 3297, 10.1039/b508541a.

35. S. Grimme, S. Ehrlich, and L. Goerigk, “Effect of the damping function in dispersion corrected density functional theory,” Journal of Computational Chemistry 32 (2011): 1456, 10.1002/jcc.21759.

36. M. J. Frisch, G. W. Trucks, H. B. Schlegel, et al., Gaussian 16, Revision A.03 (Gaussian, Inc., 2016).

37. J. N. Harvey and M. Aschi, “Modelling spin-forbidden reactions: recombination of carbon monoxide with iron tetracarbonyl,” Faraday Discussions 124 (2003): 129, 10.1039/b211871h.

38. D. S. Levine, M. Shuaibi, E. W. C. Spotte-Smith, et al., “The Open Molecules 2025 (OMol25) Dataset, Evaluations, and Models,” arXiv preprint (2025), arXiv:2505.08762, 10.48550/arXiv.2505.08762.

39. I. Batatia, D. P. Kovács, G. N. C. Simm, C. Ortner, and G. Csányi, “MACE: Higher Order Equivariant Message Passing Neural Networks for Fast and Accurate Force Fields,” Advances in Neural Information Processing Systems 35 (2022), arXiv:2206.07697.

40. J.-L. Carreón-Macedo and J. N. Harvey, “Computational study of the energetics of $^{3}Fe(CO)_4$, $^{1}Fe(CO)_4$ and $^{1}Fe(CO)_4(L)$, L = Xe, CH_4, H_2 and CO,” Physical Chemistry Chemical Physics 8 (2006): 93, 10.1039/b513325d.

41. K. Pierloot and S. Vancoillie, “Relative energy of the high-(T2g5) and low-(A1g1) spin states of [Fe(H2O)6]2+, [Fe(NH3)6]2+, and [Fe(bpy)3]2+: CASPT2 versus density functional theory,” Journal of Chemical Physics 125 (2006): 124303, 10.1063/1.2353829.

42. M. Swart and M. Gruden, “Spinning around in Transition-Metal Chemistry,” Accounts of Chemical Research 49 (2016): 2690, 10.1021/acs.accounts.6b00271.

43. M. Reiher, O. Salomon, and B. A. Hess, “Reparameterization of hybrid functionals based on energy differences of states of different multiplicity,” Theoretical Chemistry Accounts 107 (2001): 48, 10.1007/s00214-001-0300-3.

44. S. M. Bellows, T. R. Cundari, and P. L. Holland, “Spin Crossover during β-Hydride Elimination in High-Spin Iron(II)– and Cobalt(II)–Alkyl Complexes,” Organometallics 32 (2013): 4741, 10.1021/om400325x.

45. T. Takayanagi, K. Saito, H. Suzuki, Y. Watabe, and T. Fujihara, “Computational Analysis of Two-State Reactivity in β-Hydride Elimination Mechanisms of Fe(II)– and Co(II)–Alkyl Complexes Supported by β-Diketiminate Ligand,” Organometallics 38 (2019): 3582, 10.1021/acs.organomet.9b00418.

46. H. Khatua, S. Das, S. Patra, S. K. Das, S. Roy, and B. Chattopadhyay, “Iron-Catalyzed Intermolecular Amination of Benzylic C(sp^3)–H Bonds,” Journal of the American Chemical Society 144 (2022): 21858, 10.1021/jacs.2c10719.

47. X. Li, L. Dong, and Y. Liu, “Theoretical Study of Iron Porphyrin Nitrene: Formation Mechanism, Electronic Nature, and Intermolecular C–H Amination,” Inorganic Chemistry 59 (2020): 1622, 10.1021/acs.inorgchem.9b02216.

48. E. Zars, S. P. Muhm, L. Pick, et al., “Mechanistic Studies of an Iron-Catalyzed Intermolecular C–H Amination Reaction under Catalytic Conditions and Having a Large KIE,” ACS Catalysis (2026), in press, 10.1021/acscatal.5c09143.

49. M. J. Bearpark, M. A. Robb, and H. B. Schlegel, “A direct method for the location of the lowest energy point on a potential surface crossing,” Chemical Physics Letters 223 (1994): 269, 10.1016/0009-2614(94)00433-1.

50. A. T. B. Gilbert, N. A. Besley, and P. M. W. Gill, “Self-Consistent Field Calculations of Excited States Using the Maximum Overlap Method (MOM),” Journal of Physical Chemistry A 112 (2008): 13164, 10.1021/jp801738f.

51. H. Neugebauer, B. Bädorf, S. Ehlert, A. Hansen, and S. Grimme, “High-throughput screening of spin states for transition metal complexes with spin-polarized extended tight-binding methods,” Journal of Computational Chemistry 44 (2023): 2120, 10.1002/jcc.27185.

52. G. Zheng, H. A. Witek, P. Bobadova-Parvanova, et al., “Parameter Calibration of Transition-Metal Elements for the Spin-Polarized Self-Consistent-Charge Density-Functional Tight-Binding (DFTB) Method: Sc, Ti, Fe, Co, and Ni,” Journal of Chemical Theory and Computation 3 (2007): 1349, 10.1021/ct600312f.

53. O. T. Unke, S. Chmiela, M. Gastegger, K. T. Schütt, H. E. Sauceda, and K.-R. Müller, “SpookyNet: Learning force fields with electronic degrees of freedom and nonlocal effects,” Nature Communications 12 (2021): 7273, 10.1038/s41467-021-27504-0.